

1.1.1. Area of Circle

Algorithm:

Step 1: Start

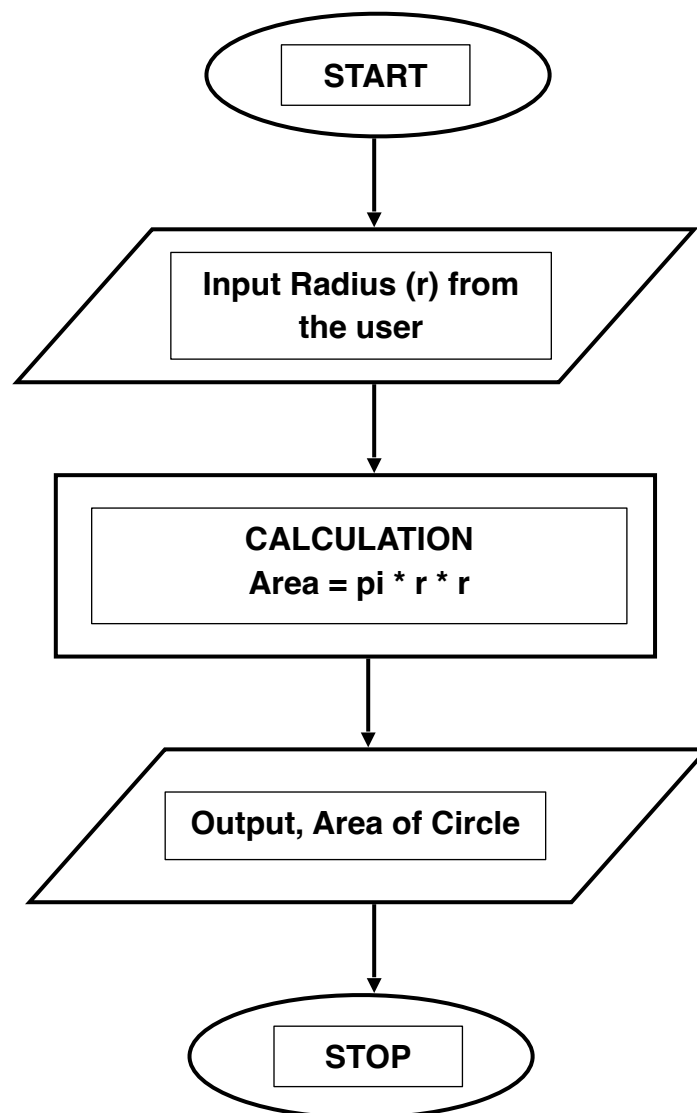
Step 3: Enter the radius of the circle.

Step 4: Calculate area using formula $\text{area} = 3.14 * r * r$

Step 5: Display the area of the circle

Step 6: Stop

Flowchart:



CODETANTRA

Home

avantika.singh.batch2025@sitnagpur.siu.edu.inSupportLogout

1.1.1. Area of Circle09:20

Write a Python program that calculates the area of a circle when the radius is provided by the user. Use $\pi = 3.14$ and display the area.

Input Format:

A single line containing a floating-point number representing the radius.

Output Format:

Print the computed area of the circle formatted to 4 decimal places.

Sample Test Cases

circlearea...

```
1
2 r = float(input())
3 a = 3.14 * r * r
4 print(f"{a:.4f}")
```

Average time0.004 s3.75 ms

Maximum time0.005 s5.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 15 ms

Expected output3.36

Actual output3.36

35.4493

Test case 23 ms

TerminalTest cases

< Prev

Reset

Submit

Next >

1.1.2. Area of Rectangle

Algorithm:

Step 1: Start .

Step 2: Input the length of the rectangle.

Step 3: Input the breadth of the rectangle.

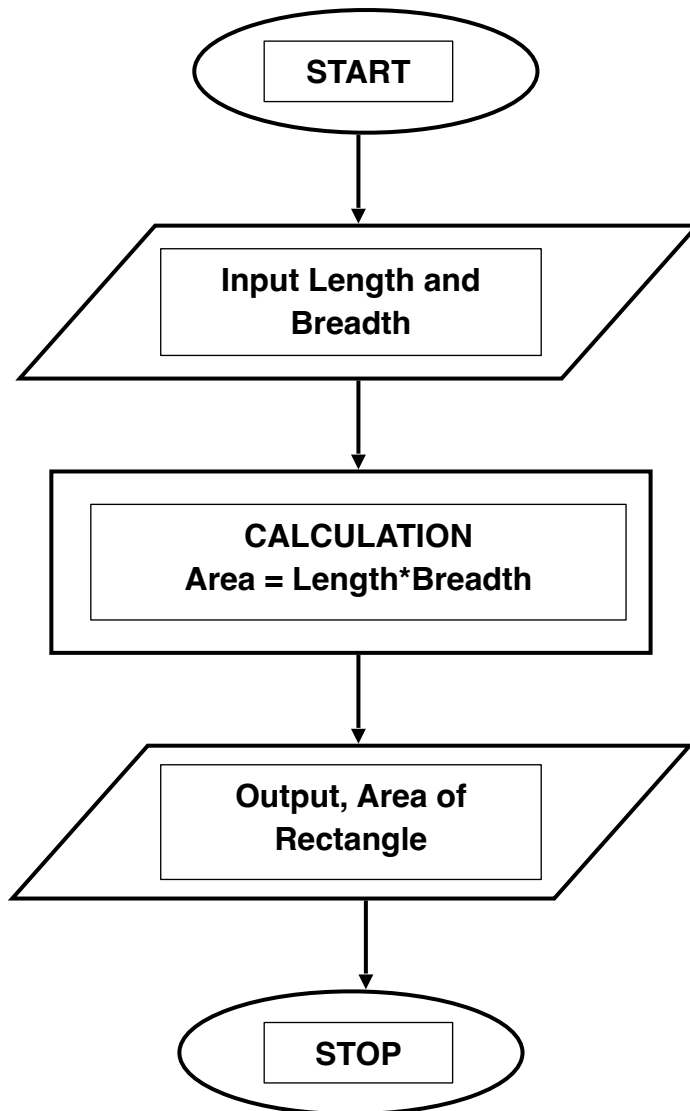
Step 4: Calculate area using formula:

$\text{area} = \text{length} \times \text{breadth}$

Step 5: Display the area of the rectangle.

Step 6: Stop .

Flowchart:



CODETANTRA

Home

avantika.singh.batch2025@sitnagpur.siu.edu.in

Support

Logout

1.1.2. Area of Rectangle

03:03

Write a Python program to calculate the area of a rectangle given its length and width.

Formula:

Area of Rectangle = Length $\times$ Width

Input Format:

- First line contains a float value representing the length of the rectangle
- Second line contains a float value representing the width of the rectangle

Output Format:

- Print the area of the rectangle as a float value formatted to 2 decimal places.

Sample Test Cases

areaOfRe...

1

l = float(input())

2

w = float(input())

3

a=l*w

4

print(f"{a:.2f}")

Average time

0.012 s

11.90 ms

Maximum time

0.021 s

21.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1 10 ms

Expected output

Actual output

10.5

10.5

5.2

5.2

54.60

54.60

Test case 2 6 ms

Terminal

Test cases

< Prev

Reset

Submit

Next >

1.1.3. Area of Square

Algorithm:

Step 1: Start.

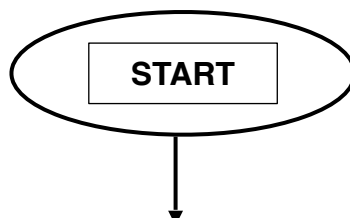
Step 2: Input the side of the square.

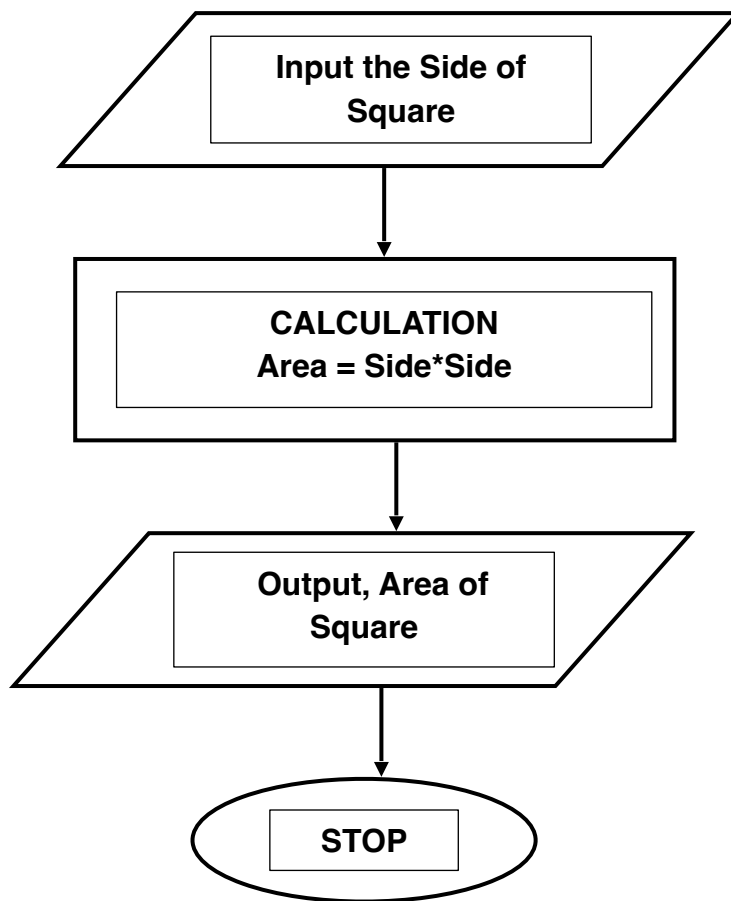
Step 3: Calculate area using formula: area = side $\times$ side

Step 4: Display the area of the square.

Step 5: Stop.

Flowchart:





CODETANTRA Home

avantika.singh.batch2025@sitnagpur.siu.edu.in Support Logout

1.1.3. Calculate Area of the Square 02:29

Write a Python program that prompts the user to enter the *side_length* of a square and computes the area of the square.

Formula:

- Area = side_length^2

Input Format:

- The input is a positive integer value that represents the *side_length* of the square.

Output Format:

- The output is a positive integer value that represents the area of the square.

Sample Test Cases +

AreaSqua...

1 side_length = int(input())
2 a=side_length*side_length
3 print(a)
4
5
6
7
8
9

Submit

Terminal Test cases

< Prev Reset Submit Next >

1.1.4 Area of Triangle

Algorithm:

Step 1: Start the program.

Step 2: Input the base of the triangle.

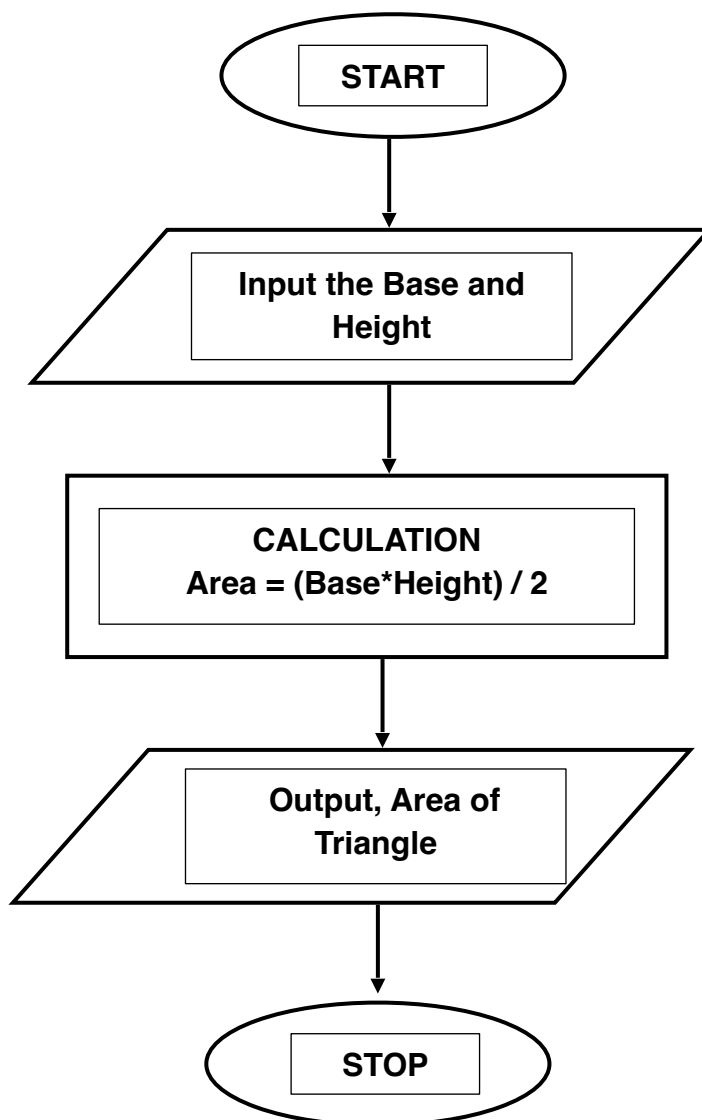
Step 3: Input the height of the triangle.

Step 4: Calculate area using formula: $\text{Area} = (\text{Base} \times \text{Height}) / 2$

Step 5: Display the area of the triangle.

Step 6: Stop the program

FLOWCHART:



[Home](#)

[sitnagpur.codetantra.com](#)

[Support](#)
[Logout](#)

1.1.4. Area of Triangle

04:13

Write a Python program that prompts the user to enter the triangle's base and height and computes the triangle's area.

Formula: $Area\ of\ Triangle = 0.5 \times base \times height$.

Input Format:

- The first line of input is the float value that represents the base of the triangle.
- The second line of input is the float value that represents the height of the triangle.

Output Format:

- The output is the floating point value that represents the area of a triangle, formatted to two decimals.

Sample Test Cases

triangleAr...

```

1 b = float(input())
2 h = float(input())
3 a=0.5*b*h
4 print(f"{a:.2f}")

```

Average time
0.007 s
7.00 ms

Maximum time
0.008 s
8.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 8 ms

Expected output	Actual output
6.54	6.54
1.23	1.23
4.02	4.02

Test case 2 6 ms

Terminal
Test cases

[< Prev](#)
[Reset](#)
[Submit](#)
[Next >](#)

1.1.5 Student Pass or Fail

Algorithm:

Step 1: Start.

Step 2: Input marks obtained by the student.

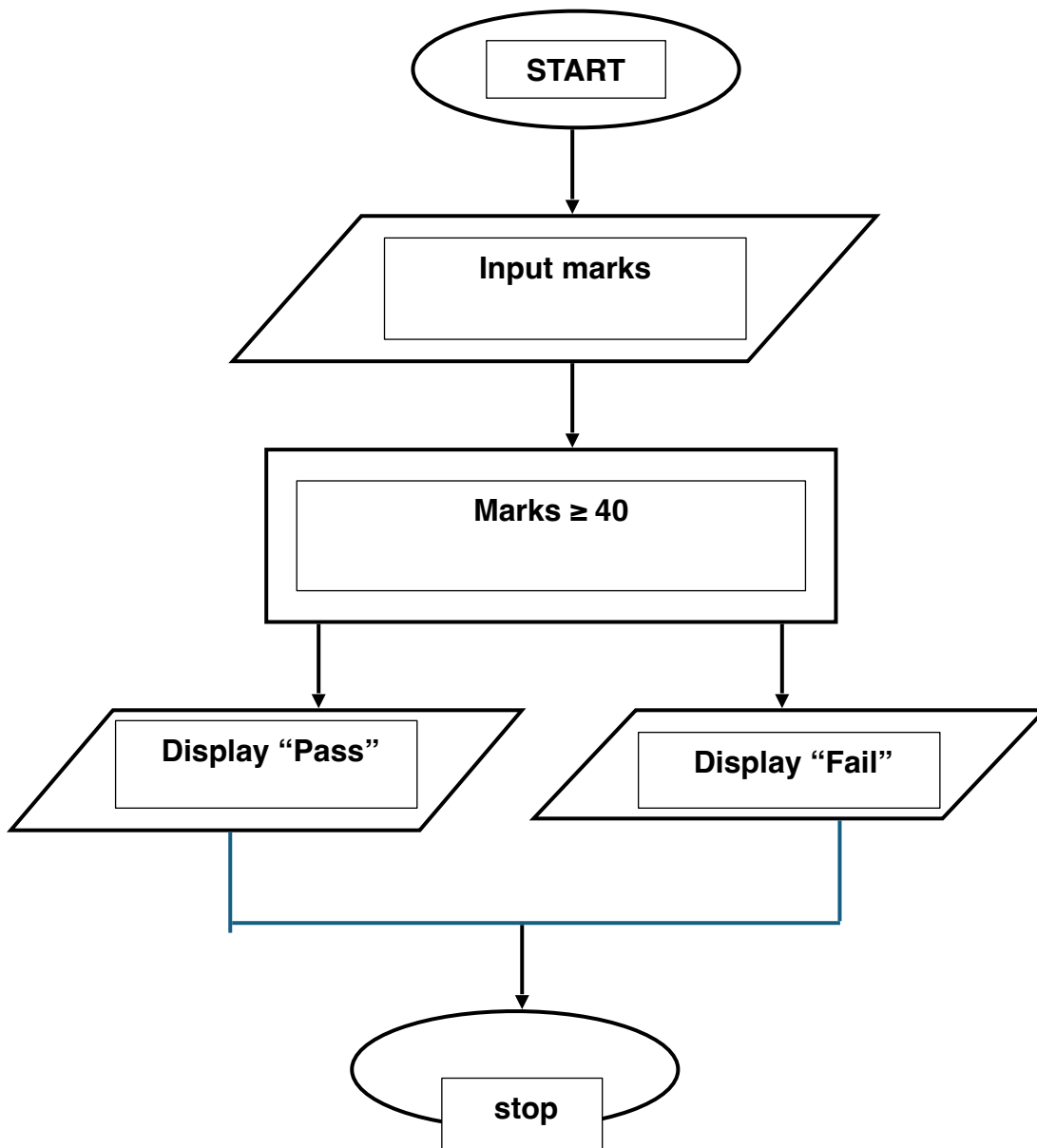
Step 3: Check if marks are greater than or equal to passing marks (for example: 40).

Step 4: If marks ≥ 40 , display “ Pass”.

Step 5: Otherwise, display “Fail”.

Step 6: Stop.

FLOWCHART:



CODETANTRA

Home

avantika.singh.batch2025@sitnagpur.siu.edu.inSupportLogout

1.1.5. Student Pass or Fail Status

01:46

Write a Python program to determine whether a student passed the exam or not based on their marks.

Pass/Fail Criteria:

- A student passes if marks ≥ 40
- A student fails if marks < 40

Input Format:

- Single line contains an integer representing the marks obtained by the student.

Output Format:

- Print "Pass" if the student passed the exam.
- Print "Fail" if the student failed the exam.

Sample Test Cases

passOrFai...

Submit

```
1 mark=int(input())
2 if mark >= 40:
3     print("Pass")
4 else:
5     print("Fail")
6
7
```

Average time0.006 s6.29 ms

Maximum time0.009 s9.00 ms

3 out of 3 shown test case(s) passed4 out of 4 hidden test case(s) passed

Test case 16 ms

Debug

Expected output45

Actual output45

Pass↵

Pass↵

Test case 26 ms

Test case 36 ms

Terminal

Test cases

< Prev

Reset

Submit

Next >

2.1.1. Roots of a Quadratic Equation

ALGORITHM

Step 1 : Start.

Step 2 : Read the coefficients a, b, and c.

Step 3 : Calculate the discriminant using:- $D=b^2-4ac$.

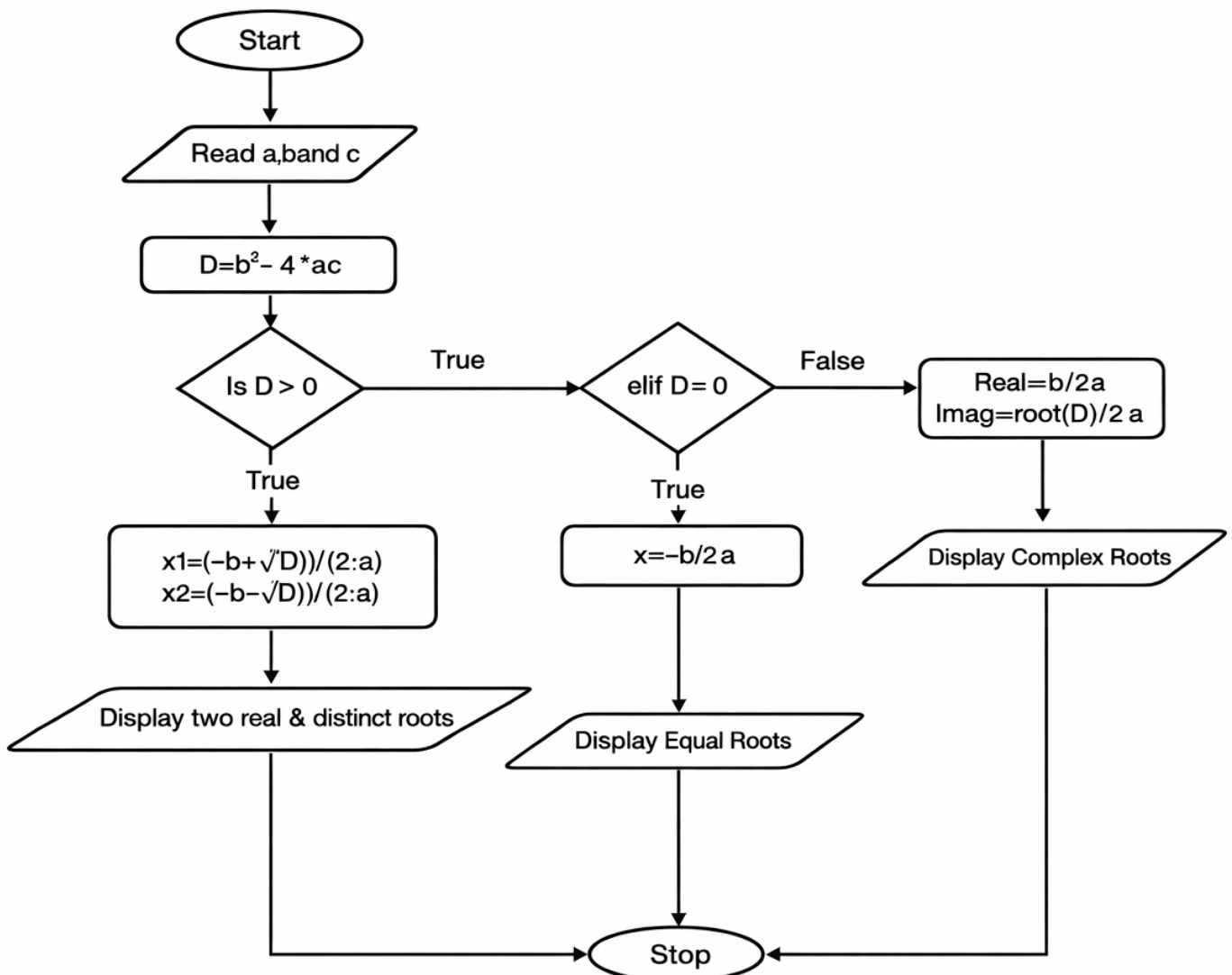
Step 4 : If $D > 0$ Compute:- $x1=(-b+\sqrt{D})/2a$ and $x2=(-b-\sqrt{D})/2a$ Display two real and distinct roots.

Step 5 : Else if $D = 0$ Compute:- $x=-b/2a$ Display equal real roots.

Step 6 : Else ($D < 0$) Compute:- Real part = $-b/2a$ Imaginary part = $\sqrt{-D}/2a$ Step 1 : Display complex roots.

Step 7 : Stop.

FLOWCHART



Course

×

+

←

→

↺

↻

sitnagpur.codetantra.com/secure/course.jsp?euclid=693fa60b79739f1e1d81ca43#/contents/693fa6c479739f1e1d81cc64/693fa76079739f1...

☆

📁

🌑

⋮

CODETANTRA

🏠 Home

avantika.singh.batch2025@sitnagpur.siu.edu.in

Support

Logout

2.1.1. Roots of a Quadratic Equation

00:25

🔍

🌙

📝

🔗

—

Write a program to find the roots of a quadratic equation, given its coefficients a , b , and c . Use the quadratic formula: $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

The discriminant $D = b^2 - 4ac$ determines the nature of the roots:

- If $D > 0$: Roots are real and different
- If $D = 0$: Roots are real and the same
- If $D < 0$: Roots are imaginary

Input Format:

- Three space-separated integers representing the coefficients a , b , and c , respectively.

Output Format:

- If roots are real and different, print:

```
root1 = <Root1>
root2 = <Root2>
```

- If roots are the same, print:

```
root1 = <Root1>
root2 = <Root2>
```

Sample Test Cases

quadratic...

Submit

```
1 a,b,c = map(int, input().split())
2
3 d = (b * b) - (4 * a * c)
4
5 if d > 0:
6     sq = d ** 0.5
7     root1 = (-b + sq) / (2 * a)
8     root2 = (-b - sq) / (2 * a)
9     print(f"root1 = {root1:.2f}")
10    print(f"root2 = {root2:.2f}")
```

Average time

0.003 s

3.33 ms

Maximum time

0.009 s

9.00 ms

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1

9 ms

Debug

⋮

📋

^

Expected output

Actual output

1 -5 6

1 -5 6

root1 = 3.00

root1 = 3.00

root2 = 2.00

root2 = 2.00

Test case 2

3 ms

⌵

Terminal

Test cases

< Prev

Reset

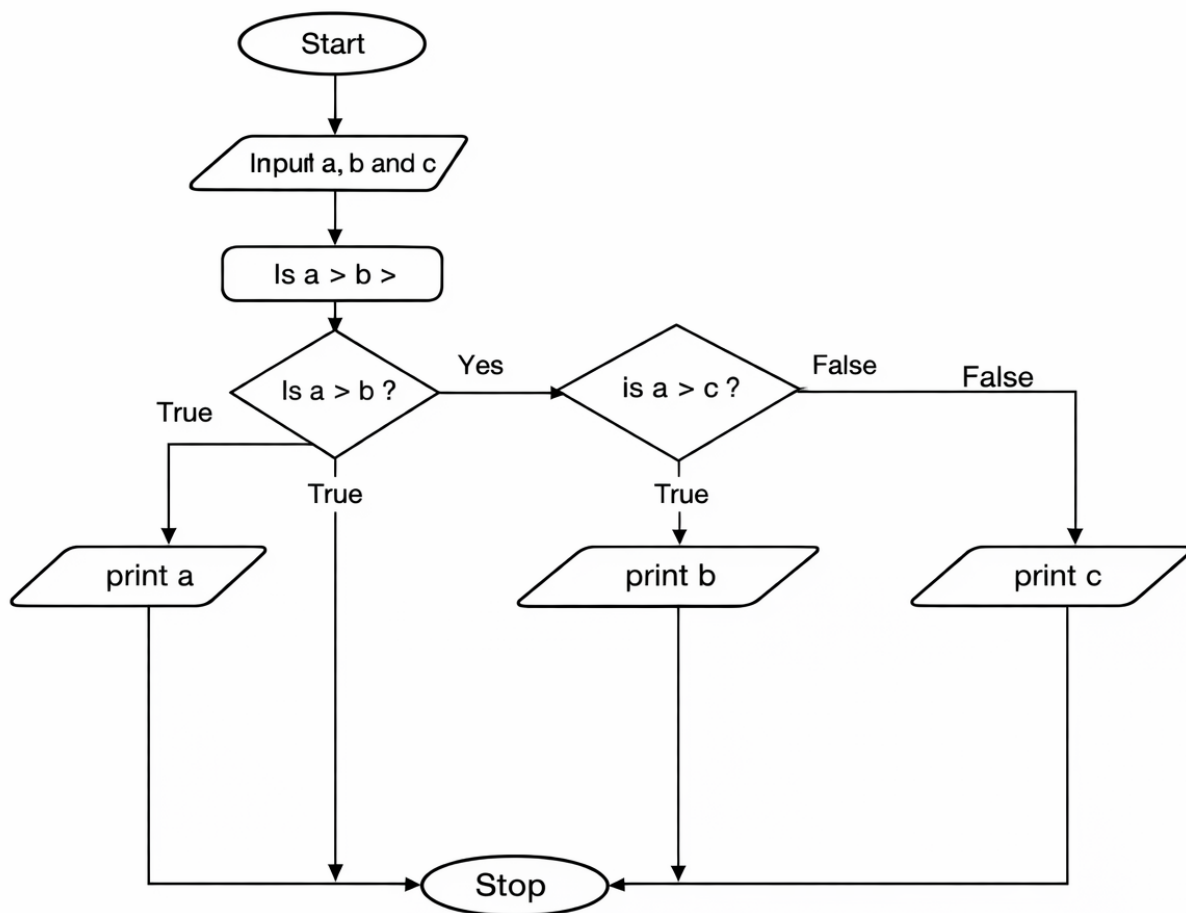
Submit

Next >

3.1.1. Largest of Three Numbers

- # 1. ALGORITHM
- 2.
 3. Step 1: Start.
 4. Step 2: Input three integers a, b, and c.
 5. Step 3: If $a \geq b$ and $a \geq c \rightarrow$ Print a as the largest number.
 6. Step 4: Else if $b \geq a$ and $b \geq c \rightarrow$ Print b as the largest number.
 7. Step 5: Else $\rightarrow$ Print c as the largest number.
 8. Step 6: Stop

FLOWCHART



Course

sitnagpur.codetantra.com/secure/course.jsp?euclid=693fa60b79739f1e1d81ca43#/contents/693fa6c879739f1e1d81cc69/693fa76379739f1...

CODETANTRA Home avantika.singh.batch2025@sitnagpur.siu.edu.in Support Logout

3.1.1. Largest of Three Numbers

00:15

Write a Python program that prompts the user to enter three integers. Print the largest of the three integers.

Input Format:

- The program will prompt the user to enter three integers, one per line.

Output Format:

- The output will display the largest integer among the three integers.

Sample Test Cases

```

1 a=int(input())
2 b=int(input())
3 c=int(input())
4 print(max(a,b,c))
5
  
```

Average time: 0.005 s (5.00 ms) | Maximum time: 0.006 s (6.00 ms)

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 (5 ms)

Expected output	Actual output
5	5
6	6
7	7
7	7

Test case 2 (6 ms)

Terminal Test cases

Prev Reset Submit Next

3.1.2. Celsius to Fahrenheit

1. ALGORITHM:

2.

3. Step 1:Start.

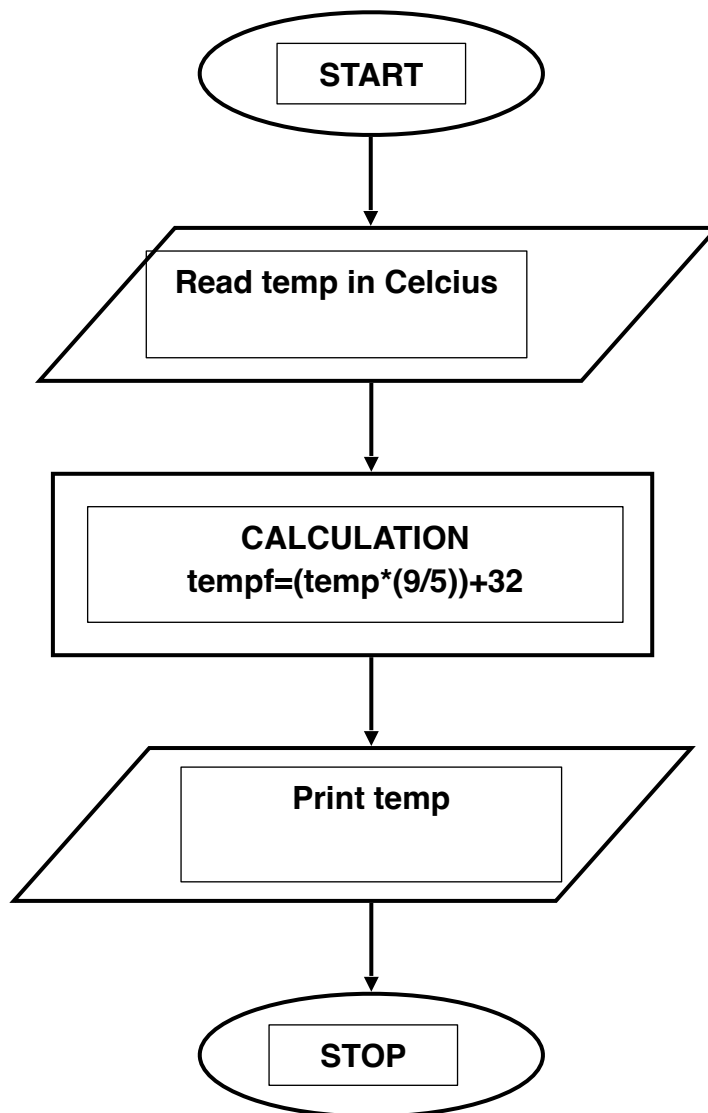
4. Step 2:Input temperature in Celsius C.

5. Step 4:Calculate Fahrenheit using the formula $F = (C \times 9/5) + 32$.

6. Step 5:Display the temperature in Fahrenheit.

7. Step 6:Stop

FLOWCHART:



Course

+

sitnagpur.codetantra.com/secure/course.jsp?euclid=693fa60b79739f1e1d81ca43#/contents/693fa6c879739f1e1d81cc69/693fa76379739f1...

☆

CODETANTRA

Home

avantika.singh.batch2025@sitnagpur.siu.edu.in

Support

Logout

3.1.2. Celsius to Fahrenheit

00:09

AA

🌙

📄

🔗

—

Write a Python program to convert temperature from Celsius to Fahrenheit.

Formula:
$$\text{Fahrenheit} = \left(\text{Celsius} \times \frac{9}{5}\right) + 32$$

Input Format:

- Single line contains a float value representing the temperature in Celsius.

Output Format:

- Print the temperature in Fahrenheit as a float value formatted to 2 decimal places.

Sample Test Cases

temperat...

Submit

1 cel=float(input())
2 f=(cel*9/5)+32
3 print(f"{f:.2f}")

Average time
0.002 s
2.38 ms

Maximum time
0.004 s
4.00 ms

4 out of 4 shown test case(s) passed
4 out of 4 hidden test case(s) passed

Test case 1 4 ms

Debug

Expected output
0.0
32.00

Actual output
0.0
32.00

Test case 2 3 ms

Test case 3 2 ms

Terminal

Test cases

< Prev

Reset

Submit

Next >